

EFFECTS OF ADVANCED WEIGHT TRAINING ON BODY-CATHEXIS AND SELF-ESTEEM^{1,2}

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Summary.—27 college students participated in an advanced weight-training course while a control group of 30 completed a physical education major theory course. Both groups were administered the Rosenberg Self-esteem Scale and the Secord and Jourard Body-cathexis Scale prior to and at the conclusion of a 16-wk. field experiment. The weight-training subjects had significantly higher self-esteem and body-cathexis scores than the control group which were attributed, in part, to significant improvement in their body composition, maximal strength, and strength endurance scores. These data support the contention that global self-esteem is a multidimensional construct responsive to positive changes in the physical self.

The importance of self-esteem, that cluster of complimentary feelings and attitudes we hold about ourselves, is recognized by theorists and researchers alike as playing a central role in any explanation of human behavior. Drawing upon the classical work of Freud, Kant, and Nietzsche, among others, Campbell (1984) has theorized that the self-esteem motive is the key to understanding and controlling human behavior. He views it as an active component in a variety of human behaviors and feeling states, including laughter, anxiety, depression, and altruism.

Research-wise, high self-esteem has been associated with consistent participation across social tasks, greater social support and acceptance from others, popularity, and self-other confidence (Ziller, 1973). Expert opinion has also suggested that high self-esteem is positively correlated with such personality traits as assertiveness, willingness to take risks, happiness, independence, and openness to personal growth (Gaudin, 1989).

The California Task Force to Promote Self-esteem and Personal and Social Responsibility, the first state commission in the United States to study the relationship between self-esteem and social problems, has taken the position that low self-esteem is associated with alcoholism, drug abuse, crime and violence, child abuse, teenage pregnancy, prostitution, chronic welfare dependency, and failure of children to learn. While the commission's assumptions may be somewhat overzealous, confusing in some cases, cause with

¹The authors acknowledge the contributions of Dr. Ravi Shukla, State University of New York, College at Brockport, to the conceptualization of the research problem and the statistical analysis of the data.

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effect (e.g., does low self-esteem cause juvenile delinquency or does juvenile delinquency cause low self-esteem?; see McCarthy & Hoge, 1984), there does seem sufficient evidence to support the proposition that self-esteem may well be one of the most precious of human assets. Perhaps that is why "... much of our behavior, whether conscious or not, is directed toward maximizing our chances of feeling good about ourselves" (Fox, 1988, p. 231). Fox even suggests that self-esteem is the most accessible index of emotional adjustment and mental health.

An individual's perception of self-worth can be viewed as a fluctuating self-attitude which rises and falls as a result of changing roles, expectations, performances, and other situational characteristics (Demo, 1985). This "state approach" to self-esteem contrasts sharply with the "trait approach" wherein self-esteem is viewed as a personality trait characterized by considerable stability from one situation to the next (Jackson & Paunonen referenced by O'Malley & Bachman, 1983).

A difference of opinion also surrounds the dimensionality of this evaluative component of the self-concept. Is self-esteem a unidimensional or multidimensional construct? Fox (1988) has argued that individuals can have very different evaluative perceptions of themselves in separate aspects of their lives, e.g., social relationships, academic matters, physical appearance, etc. According to Fox, some hierarchical model makes the most theoretical sense, that is, a structure that includes a measure of global self-esteem at the apex, acting as a superordinate outcome of perceptions of self in a variety of domains.

If self-esteem is a multidimensional construct, then surely one of the specific domains that contributes to its valuation is the physical self. Interestingly, the task of identifying the most significant components of the physical self-esteem structure has only been recently undertaken. Fox and Corbin (1989) have speculated that such a structure consists of five components, namely, sports competence, attractiveness of the body, physical strength, physical condition/exercise, and a general type of physical self-worth.

Sonstroem (1984) reviewed 16 studies on the effects of exercise on self-esteem and concluded that the consistency of positive results suggested a basis for belief in the salutary effects of physical training programs. However, given some methodological problems, the reviewer could not say why or in what manner exercise programs affect self-esteem or which types of participants are most responsive to change.

Of direct relevance to the present study was Secord and Jourard's (1953) finding that feelings about the body are commensurate with feelings about the self. Specifically, they found that low body-cathexis was associated with anxiety and insecurity. More recently, Wendel and Lester (1988) reported positive correlations between subjective evaluations of waist, hips, weight,

and self-esteem for college women and positive correlations between voice, resistance to illness, physical profile, and self-esteem for college men.

Although not directly applicable to the present study, researchers have also reported significant positive correlations between self-esteem and sports participation (Brennan, 1985; Guyot, Fairchild, & Hill, 1981; Hines & Groves, 1989; Smith, 1986), athletic ability (Hollinger & Fleming, 1985), father's interest in sports (Felker, 1968; Kay, Felker, & Varoz, 1972), and parent/child differences in judgments of sports ability (McElroy & Kirkendall, 1981).

While the effects of a variety of exercise and fitness training programs on self-esteem have been investigated (Brown, Morrow, & Livingston, 1982; Hilyer & Mitchell, 1979), only a few studies have looked at weight-training programs (Trujillo, 1983; Tucker, 1982) and none have studied subjects who trained beyond the beginning level and for intervals longer than 10 weeks. The present study extends the research literature by investigating the effects of a 16-wk. advanced weight-training program on measures of body-cathexis and global self-esteem. Whether global self-esteem is a fluctuating personality trait sensitive to changes in other dimensions of the total self, e.g., the physical domain, is a research question that has important mental health implications (Anshel, Muller, & Owens, 1986; Walker & Greene, 1986).

METHOD

Subjects

The trained group consisted of 27 American-born Caucasian college students (17 men, 10 women) enrolled fall semester, 1989 in an advanced weight-training course. The mean age of the subjects was 23.6 yr. ($SD = 4.8$). The group met three times a week for 1-hour workouts for a total of 16 weeks. The subjects received instruction in the practical and theoretical aspects of strength training. Individual workout programs consisted of various combinations of free weight, Nautilus, Universal, and calisthenic exercises.

The control group consisted of 30 American-born Caucasian college students (16 men, 14 women) enrolled in a theory course entitled "Physiological Perspective of Sport and Exercise" designed for physical education majors. The mean age of the group was 23.9 yr. ($SD = 4.9$). The subjects met twice a week for a 1-hour lecture for 16 weeks. The course content included information about physiological responses of the human body during exercise and physical activity and the physiology of fitness. The course was taught by the same instructor who taught the advanced weight-training class.

Procedure

During the second and sixteenth week of the semester, the students in both courses completed two paper-and-pencil "self" questionnaires, the Rosen-

berg Self-esteem Scale (1965) and the Secord and Jourard Body-cathexis Scale (1953). The former is a 10-item, self-report scale of global attitudes toward the self. The scale has five positively and five negatively worded items (the scoring is reversed for the latter). Using a four-point Likert response format, the subject responds: (1) agree very much, (2) agree, (3) don't agree, and (4) don't agree at all. The lower the total score for the 10 items, the higher the subject's self-esteem. Scores can theoretically range from 10 (very high self-esteem) to 40 (very low self-esteem). Rosenberg (1979) reported a coefficient of reproducibility of 92%, a coefficient of scalability of 72%, and a test-retest reliability of 0.88. Goldsmith (1986) observed that the scale seems to measure a single or global dimension of attitude toward the self with age-homogeneous populations.

The Body-cathexis Scale is a 46-item, five-choice instrument which appraises the feelings of an individual towards the various parts and functions of his own body. Specifically, subjects are asked whether they have positive or negative feelings about particular body parts, e.g., chest, and bodily functions, e.g., sleeping. A total Body-cathexis score is obtained by summing the ratings for each of the 46 body items and dividing by 46. The subject's possible responses were: (1) have strong feelings and wish change could somehow be made, (2) don't like, but can put up with, (3) have no particular feelings one way or the other, (4) am satisfied, and (5) consider myself fortunate. The higher the total average score, the more satisfied the subject feels about his body parts and bodily processes. Scores can theoretically range from an average of 1 (very dissatisfied) to 5 (very satisfied). The reported split-half reliability of the scale is 0.81 (Secord & Jourard, 1953).

At the start of the third week and again during the sixteenth or final week of the course, all subjects in the advanced weight-training course were tested for strength, body composition, endurance, and flexibility. Maximal strength was assessed using one-repetition maximum (1RM) bench press (pectoralis and arm extensor muscle groups) and leg press (quadriceps and hip flexor muscle groups) tests. Upper and lower body strength endurance was measured via bench press (maximum number of repetitions using 50% of the 1RM weight) and leg press endurance tests (maximum number of repetitions using 50% of the 1RM weight), respectively. Abdominal strength and endurance was tested by a 1-min. speed sit-up test. Percent body fat was determined using equations developed by Jackson and Pollock (1978) and Jackson, Pollock, and Ward (1980). Extent flexibility was determined using sit-and-reach and shoulder flexion tests.

The statistical significance ($p < .05$) of inter- and intragroup mean differences in self-esteem and body-cathexis was estimated by the t statistic for independent sample means. Changes in maximal strength, strength endurance, and flexibility were statistically analyzed using a multivariate analysis

of variance. Changes between pre- and posttest body composition measures (i.e., skinfold sites and percent body fat) were analyzed using the t statistic for correlated sample means.

RESULTS

Rosenberg Self-esteem Scores: Between Groups

There was no significant difference between the two groups on the Rosenberg Self-esteem Scale prior to the start of the weight-training program ($t_{35} = -1.13$); see Table 1. An analysis of posttest mean scores showed a significant difference in favor of the trained group ($t_{35} = -3.15$, $p < .05$). Specifically, the weight-training group scored significantly better on five of the ten Rosenberg items. They were more satisfied and pleased with themselves, felt more important, and had better feelings about themselves than the subjects in the control group. They were also less likely to rate themselves as useless or think of themselves as bad.

TABLE 1
GROUP MEANS AND STANDARD DEVIATIONS FOR ROSENBERG SELF-ESTEEM
AND SECORD AND JOURARD BODY-CATHEXIS SCALES

Group		Rosenberg		Secord and Jourard	
		Pretest	Posttest	Pretest	Posttest
Experimental, $n = 27$	M	1.77	1.59	3.53	3.69
	SD	0.45	0.40	0.41	0.38
Control, $n = 30$	M	1.92	1.97	3.38	3.19
	SD	0.52	0.50	0.50	0.44

Rosenberg Self-esteem Scale: Within Groups

The within group analysis of pre- and posttest Rosenberg scores for the weight-training group was not statistically significant ($t_{25} = -1.57$). Analysis of individual questions did show significant improvement on two items, namely, Item 2, "In general, I am satisfied and pleased with myself" ($t_{25} = -2.51$, $p < .05$), and Item 3, "Sometimes I feel useless" ($t_{25} = -2.08$, $p < .05$). There was no significant change between pre- and posttest over-all or on individual item scores for the control group.

Body-cathexis Scale: Between Groups

Pretreatment differences between the trained and control groups were discovered for only 4 or 8.77% of the 46 Secord and Jourard body-cathexis questionnaire items. Specifically, control group subjects were more satisfied with their breathing while the trained group was more satisfied with their head shape, eyes, and lips. Both groups were similarly satisfied with their body image prior to the start of the weight-training program ($t_{35} = 1.23$).

Comparison of the posttreatment group scores yielded a significant dif-

ference in favor of the weight-training group ($t_{35} = 3.58, p < .05$); see Table 1. Specifically, the weight trainers scored significantly higher on 24 or 52.2% of the items. They were more satisfied with their appetite, elimination, waist, energy level, breathing, back, amount of exercise, arms, legs, sleeping, body build, posture, profile, chest, health, and weight.

Body-cathexis Scale: Within Groups

There was no significant difference between the weight-training group's pre- and posttest over-all scores ($t_{16} = 1.55$); see Table 1. There was, however, significant body-cathexis improvement in four specific areas, namely, energy level ($t_{25} = 2.64, p < .05$), amount of exercise ($t_{25} = 2.05, p < .05$), back ($t_{25} = 2.61, p < .05$), and elimination ($t_{25} = 2.73, p < .05$). As expected, there was no significant difference between the control group's pre- and posttest scores after 16 weeks of classroom activity ($t_{28} = -1.56$). In fact, there was a nonsignificant decrease in their over-all body-cathexis score; see Table 1.

TABLE 2
EXPERIMENTAL GROUP MEANS AND STANDARD DEVIATIONS FOR
BODY COMPOSITION MEASURES

Site*		Pretest	Posttest
Women			
Triceps	<i>M</i>	14.91	13.56
	<i>SD</i>	4.31	3.98
Suprailium	<i>M</i>	11.73	10.50
	<i>SD</i>	5.38	5.56
Thigh	<i>M</i>	20.81	19.94
	<i>SD</i>	5.81	4.80
Percent Body Fat	<i>M</i>	19.58	17.98
	<i>SD</i>	5.55	5.03
Men			
Pectoralis	<i>M</i>	11.32	10.79
	<i>SD</i>	4.21	4.64
Abdomen	<i>M</i>	17.31	15.06
	<i>SD</i>	6.33	5.93
Thigh	<i>M</i>	12.71	11.50
	<i>SD</i>	4.91	4.28
Percent Body Fat	<i>M</i>	11.73	10.52
	<i>SD</i>	3.96	4.07

*All skinfold site measurements in millimeters.

Body Composition

For female weight trainers, there were no significant differences between the pre- and posttest scores for the three skinfold measures (triceps,

suprailium, thigh) but they did experience a significant reduction in body fat, from 19.58% to 17.98% ($t_7 = -2.64$, $p < .05$). The skinfold results were not totally unexpected because these female weight trainers were relatively lean prior to the start of the program. The fact that they were able to reduce their skinfold scores 9.1% for the triceps, 10.5% for the suprailium, and 4.2% for the thigh reflects well on how hard they exercised.

For male weight trainers, there was no significant difference between their pre- and posttest measures for the pectoralis (chest) skinfold site. There was a marginally significant reduction for the thigh skinfold site ($t_{16} = -2.10$, $p < .06$) and a significant reduction at the abdominal site ($t_{16} = -2.99$, $p < .05$). While these subjects reduced their percent body fat 10.3% on average, it was not statistically significant ($t_{16} = -1.98$); see Table 2.

Maximum Strength and Strength Endurance

All subjects who weight trained significantly improved their scores in the 1RM bench press ($F_{1,25} = 55.97$, $p < .05$), 1RM leg press ($F_{1,25} = 150.46$, $p < .05$), bench-press endurance test ($F_{1,25} = 180.49$, $p < .05$), leg-press endurance test ($F_{1,25} = 4.88$, $p < .05$), and 1-min. speed sit-up test ($F_{1,25} = 20.64$, $p < .05$); see Table 3.

TABLE 3
TRAINED GROUP MEANS AND STANDARD DEVIATIONS FOR
MAXIMAL STRENGTH, STRENGTH ENDURANCE, AND FLEXIBILITY

Test		Women		Men	
		Pretest	Posttest	Pretest	Posttest
Bench press, one-repetition maximum (lbs.)	<i>M</i>	80.91	117.27	225.00	246.00
	<i>SD</i>	13.75	19.41	46.79	42.39
Leg press, one-repetition maximum (lbs.)	<i>M</i>	177.78	351.11	502.50	581.56
	<i>SD</i>	73.45	86.81	152.12	119.17
Bench press endurance, 50% of 1RM (repetitions)	<i>M</i>	25.36	34.36	27.44	34.63
	<i>SD</i>	5.87	7.95	7.75	2.83
Leg press endurance, 50% of 1RM (repetitions)	<i>M</i>	37.71	49.00	26.73	38.73
	<i>SD</i>	18.93	16.85	11.05	4.30
1-min. speed sit-ups (repetitions)	<i>M</i>	43.00	62.00	51.35	72.53
	<i>SD</i>	8.65	6.40	8.10	7.72
Sit-and-reach (cm)	<i>M</i>	41.06	44.00	35.24	37.76
	<i>SD</i>	5.26	3.77	5.88	5.51
Shoulder flexion (inches)	<i>M</i>	14.82	16.87	13.00	15.29
	<i>SD</i>	3.31	1.76	1.66	2.39

Extent Flexibility

While the scores for the shoulder flexion and sit-and-reach tests improved between the pre- and posttests for both men and women, the

differences were not significant; see Table 3. Of special interest, from a biomechanical point of view, was the fact that increases in maximal strength, as previously noted, did not result in reduced flexibility in the joint segments tested.

DISCUSSION

The fact that the experimental group had significantly higher body-cathexis scores than the control group on 24 of the 46 body-cathexis items following 16 weeks of advanced weight training was certainly not surprising. Their hard work in the weight-training room, confirmed by improved body composition, maximal strength, and muscular endurance scores, was associated with more positive evaluations of several body parts and bodily functions. Failure to discover significant improvement in their over-all body-cathexis score was not unexpected. There was little reason to believe that their individual weight-training programs would result in more positive feelings about their hair, facial complexion, hands, distribution of hair, nose, fingers, ears, chin, shape of head, eyes, lips, teeth, forehead, and feet. On the other hand, significant improvement was expected and noted in such key health-related areas as energy level, frequency of exercise, and elimination. It seems reasonable to conclude that the advanced weight-training course did have positive effects on the subjects' self-perceptions of physical attractiveness, physical condition/exercise, and physical self-worth, three key components of the physical self-esteem structure.

While this study provides evidence to support the proposition that physiological and anthropometric changes did enhance body image, is there also reason to believe that the weight trainers' global self-esteem was also improved? Recall that the Rosenberg self-esteem scores for the control group did not change over the 16-wk. test period while the subjects who weight-trained reported they felt more satisfied and pleased with themselves and less likely to view themselves as useless. Although there was no significant difference between the experimental group's pre- and posttest over-all self-esteem scores, the group change score was in the anticipated (positive) direction.

While these data may not be persuasive enough to state categorically that the advanced weight-training program significantly improved global self-esteem, they do provide inferential support for the popular contention that body image is related to self-esteem (Fisher, 1986; Fox, 1988; Secord & Jourard, 1953; Wendel & Lester, 1988). It is reasonable to argue that the positive body composition, strength, and endurance changes associated with the weight-training program enhanced key components of the subjects' physical self which, in turn, contributed to an improvement in selected global self-esteem scores. If this is what occurred, then it lends support to the belief that self-esteem is a multidimensional, situation-specific construct.

It is important to note that, while investigations such as this one, conducted in an applied, naturalistic setting, can provide potentially valuable information, the methodological shortcomings must also be recognized (Sonstroem, 1984). First, convenience samples were used as opposed to a random sample or the random assignment of subjects to groups. Second, since every effort was made to guarantee the anonymity of the subjects in both groups, no attempt was made to pair the subjects' pre- and posttest scores on the body-cathexis and self-esteem scales, only group data were analyzed. This decision ruled out the use of the recommended analysis of covariance for comparing posttest mean differences following pretest differences. However, this limitation was largely overcome in a subsequent analysis which compared the change scores for the two groups. Both the Rosenberg ($t_{35} = -1.83$, $p < .05$) and Secord and Jourard ($t_{35} = 3.12$, $p < .05$) change-score comparisons significantly favored the weight-training group. This second analysis provides additional statistical evidence that the advanced weight-training program positively influenced the body-cathexis and global self-esteem scores of the weight trainers.

Third, it must also be pointed out that no attempt was made to identify the specific agents in the weight-training program that might have contributed to the subjects' global self-esteem. It is possible that the improved self-esteem scores had less to do with improvement in fitness and body composition and more to do with the special social experiences subjects shared in the weight-training facility, their close identification with a caring, adult leader, or the reinforcement and support they received from significant others both inside and outside of class (Jasnoski, Holmes, Solomon, & Aguiar, 1981).

Lastly, while these findings add to our understanding of the potential psychological benefits of formal exercise programs in a generic sense, the reader is reminded that no attempt was made to examine individual subjects' data. For example, it is not known whether the psychological benefits of exercise programs are equally available to all participants, regardless of initial levels of physical fitness, physical self-worth, and global self-esteem.

While these data do not explicitly establish a statistical connection between the advanced weight-training program and global self-esteem improvement, the inference is heavy that there exists a nexus among heavy resistance exercise programs, body-cathexis, and self-esteem. It seems logical to assume that significant improvements in maximal strength, muscular endurance, and body composition can lead to feelings of greater satisfaction with the physical self which, in turn, can have a positive effect on general measures of self-evaluation. We would hope that this initial effort at illuminating these promising relationships will spur further investigation.

REFERENCES

- ANSHEL, M. H., MULLER, D., & OWENS, V. L. (1986) Effect of a sports camp experience on the multidimensional self-concepts of boys. *Perceptual and Motor Skills*, 63, 363-366.
- BRENNAN, A. (1985) Participation and self-esteem: a test of six alternative explanations. *Adolescence*, 20, 445-466.
- BROWN, E. Y., MORROW, J. R., & LIVINGSTON, S. M. (1982) Self-concept changes in women as a result of training. *Journal of Sport Psychology*, 4, 354-363.
- CAMPBELL, R. (1984) *The new science: self-esteem psychology*. Lanham, MD: Univer. Press of America.
- CRANDALL, R. (1973) The measurement of self-esteem and related concepts. In J. P. Robinson & P. R. Shaver (Eds.), *Measures of psychological attitudes*. Ann Arbor, MI: Univer. of Michigan Press. Pp. 45-167.
- DEMO, D. H. (1985) The measurement of self-esteem: refining our methods. *Journal of Personality and Social Psychology*, 48, 1490-1502.
- FELKER, D. W. (1968) Relationships between self-concept, body build, and perception of fathers' interest in sports in boys. *Research Quarterly*, 39, 513-517.
- FISHER, S. (1986) *Development and structure of the body image*. Hillsdale, NJ: Erlbaum.
- FOX, K. R. (1988) The self-esteem complex and youth fitness. *Quest*, 40, 230-246.
- FOX, K. R., & CORBIN, C. B. (1989) The physical self-perception profile: development and preliminary validation. *Journal of Sport & Exercise Psychology*, 11, 408-430.
- GAUDIN, K. (1989) Self-esteem isn't a dirty word. *Democrat and Chronicle* (Rochester, NY), March 7, 1989, p. 50.
- GOLDSMITH, R. E. (1986) Dimensionality of the Rosenberg Self-esteem Scale. *Journal of Social Behavior and Personality*, 1, 253-264.
- GUYOT, G. W., FAIRCHILD, L., & HILL, M. (1981) Physical fitness, sport participation, body build, and self-concept of elementary school children. *International Journal of Sport Psychology*, 12, 105-116.
- HILYER, J. C., & MITCHELL, W. (1979) Effect of physical fitness training combined with counseling on the self-concept of college students. *Journal of Counseling Psychology*, 26, 427-436.
- HINES, S., & GROVES, D. L. (1989) Sports competition and its influence on self-esteem development. *Adolescence*, 24, 861-869.
- HOLLINGER, C. L., & FLEMING, E. S. (1985) Social orientation and the social self-esteem of gifted and talented female adolescents. *Journal of Youth and Adolescence*, 14, 389-399.
- JACKSON, A. S., & POLLOCK, M. L. (1978) Generalized equations for predicting body density of men. *British Journal of Nutrition*, 40, 497.
- JACKSON, A. S., POLLOCK, M. L., & WARD, A. (1980) Generalized equations for predicting body density of women. *Medicine and Science in Sports and Exercise*, 12, 175-182.
- JASNOSKI, M. L., HOLMES, D. S., SOLOMON, S., & AGUIAR, C. (1981) Exercise, changes in aerobic capacity, and changes in self-perceptions: an experimental investigation. *Journal of Research in Personality*, 15, 460-466.
- KAY, R. S., FELKER, D. W., & VAROZ, R. O. (1972) Sports interests and abilities as contributors to self-concept in junior high school boys. *Research Quarterly*, 43, 208-215.
- MCCARTHY, J. D., & HOGE, D. R. (1984) The dynamics of self-esteem and delinquency. *American Journal of Sociology*, 90, 396-410.
- MCELROY, M. A., & KIRKENDALL, D. R. (1981) Conflict in perceived parent/child sport ability judgments. *Journal of Sport Psychology*, 3, 244-247.
- O'MALLEY, P. M., & BACHMAN, J. G. (1983) Self-esteem: change and stability between ages 13 and 23. *Developmental Psychology*, 19, 257-268.
- ROSENBERG, M. (1965) *Society and the adolescent self-image*. Princeton, NJ: Princeton Univer. Press.
- ROSENBERG, M. (1979) *The conceiving self*. New York: Basic Books.
- SECORD, P. F., & JOURARD, S. M. (1953) The appraisal of body-cathexis: body-cathexis and the self. *Journal of Consulting Psychology*, 17, 343-347.
- SMITH, T. L. (1986) Self-concepts of youth sport participants and nonparticipants in Grades 3 and 6. *Perceptual and Motor Skills*, 62, 863-866.

- SONSTROEM, R. J. (1984) Exercise and self-esteem. In R. L. Terjung (Ed.), *Exercise and sport sciences reviews*. Vol. 11. Lexington, MA: Collamore Press. Pp. 123-155.
- TRUJILLO, C. M. (1983) The effect of weight training and running exercise intervention programs on the self-esteem of college women. *International Journal of Sport Psychology*, 14, 162-173.
- TUCKER, L. A. (1982) Effect of a weight-training program on the self-concepts of college males. *Perceptual and Motor Skills*, 54, 1055-1061.
- WALKER, L. S., & GREENE, J. W. (1986) The social context of adolescent self-esteem. *Journal of Youth and Adolescence*, 15, 315-322.
- WENDEL, G., & LESTER, D. (1988) Body-cathexis and self-esteem. *Perceptual and Motor Skills*, 67, 538.
- ZILLER, R. C. (1973) *The social self*. New York: Pergamon.

Accepted June 17, 1991.